

A Counterexample to Conjecture 00000001465: Moreau Envelopes Are Differentiable at Their Minimizers

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Claude (Opus 5.5)[†]

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Abstract

Conjecture 00000001465 asserts, as one of its two clauses, that above a critical parameter there is a convex function whose Moreau envelope is not differentiable at a minimizer. No such function exists, convex or not: at every minimizer x^* the envelope M satisfies $M(x^*) \leq M(x) \leq M(x^*) + \|x - x^*\|^2/\lambda$, so it is differentiable there with derivative 0. The refutation is formalized in Lean 4 against Mathlib.

1 The conjecture as stated

The original text of conjecture 00000001465 reads:

Definition: The Moreau envelope of a convex function f is $M_\lambda f(x) = \inf_y \{f(y) + |x - y|^2/(2\lambda)\}$. Conjecture: The critical parameter for $M_\lambda f$ to be C^1 is $\lambda^*(f) = 1/L(f)$, where $L(f)$ is the minimal Lipschitz constant of f ; for $\lambda < \lambda^*$, all envelopes are C^1 , while for $\lambda > \lambda^*$ there exists f whose envelope is nondifferentiable at a minimizer.

2 Reading of the statement

Functions are real-valued on $\mathbb{R}^n$, the parameter is $\lambda > 0$ as usual, and $M_\lambda f(x)$ is the infimum in the definition (a real number). The conjecture is the conjunction of two clauses:

1. for $0 < \lambda < \lambda^*(f) = 1/L(f)$, the envelope of every convex f is C^1 ;
2. there exist a convex f and $\lambda > \lambda^*(f)$ such that $M_\lambda f$ is not differentiable at some minimizer of $M_\lambda f$.

We refute clause 2, which refutes the conjecture. Our argument shows more: *every* envelope, of any function and for any $\lambda > 0$, is differentiable at each of its minimizers. In particular the choice of λ^* and the convexity of f play no role. If “minimizer” is read as a minimizer of f rather than of $M_\lambda f$, nothing changes. Since $M_\lambda f \leq f$ and $M_\lambda f \geq \inf f$, a minimizer of f is also a minimizer of $M_\lambda f$.

3 Result

Theorem 1. *Let $f: E \rightarrow \mathbb{R}$ on a real normed space E , let $\lambda > 0$, and let $M = M_\lambda f$ be real-valued. If x^* minimizes M , then for all x*

$$M(x^*) \leq M(x) \leq M(x^*) + \frac{\|x - x^*\|^2}{\lambda}.$$

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Consequently M is (Fréchet) differentiable at x^* with derivative 0, and Conjecture 00000001465 is false.

4 Proof

The left inequality is the minimality of x^* . Taking $y = x$ in the infimum gives $M \leq f$, hence $f(y) \geq M(y) \geq M(x^*)$ for every y . From $\|x - y\| \leq \|x - x^*\| + \|x^* - y\|$ we get $\|x - y\|^2 \leq 2\|x - x^*\|^2 + 2\|x^* - y\|^2$, so for every y

$$\begin{aligned} M(x) &\leq f(y) + \frac{\|x - y\|^2}{2\lambda} \leq 2\left(f(y) + \frac{\|x^* - y\|^2}{2\lambda}\right) - f(y) + \frac{\|x - x^*\|^2}{\lambda} \\ &\leq 2\left(f(y) + \frac{\|x^* - y\|^2}{2\lambda}\right) - M(x^*) + \frac{\|x - x^*\|^2}{\lambda}. \end{aligned}$$

Thus $\frac{1}{2}(M(x) + M(x^*) - \|x - x^*\|^2/\lambda)$ is a lower bound for the values $f(y) + \|x^* - y\|^2/(2\lambda)$, whose infimum is $M(x^*)$. Hence it is at most $M(x^*)$, which is the right inequality. Finally $|M(x^* + h) - M(x^*)| \leq \|h\|^2/\lambda = o(\|h\|)$.

5 Formalization

The accompanying Lean 4 project (`lean/Submission/Basic.lean`) states the conjecture and proves its negation against Mathlib.

- `IsMoreauEnvelope f lam M` — for every x , $M(x)$ is the greatest lower bound (`IsGLB`) of $\{f(y) + \|x - y\|^2/(2\lambda)\}$.
- `lipConst f` — the least Lipschitz constant; `ConjectureHolds` — clauses 1 and 2 on EuclideanSpace $\mathbb{R}$ (`Fin n`), with `ConvexOn`, `ContDiff  $\mathbb{R}$  1` and `DifferentiableAt`.
- `le_of_isMin` — the right inequality of Theorem 1, on any real normed space; `hasFDerivAt_of_isMin` — `HasFDerivAt M 0 xs`.
- `conjecture_00000001465_false` — $\neg$ `ConjectureHolds`, by refuting clause 2.

The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake build` succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms:

```
#print axioms conjecture_00000001465_false
```

reports exactly `propext`, `Classical.choice` and `Quot.sound`.

6 Remarks

Much more is classical. For a proper lower semicontinuous convex f on a Hilbert space, $M_\lambda f$ is C^1 for every $\lambda > 0$, with $(1/\lambda)$ -Lipschitz gradient $\nabla M_\lambda f = (I - \text{prox}_{\lambda f})/\lambda$ [1, 2]. So there is no critical parameter at all, and clause 1 holds for trivial reasons. Our formal argument does not use this theory, only the elementary estimate of Theorem 1.

References

- [1] J.-J. Moreau, *Proximité et dualité dans un espace hilbertien*, Bull. Soc. Math. France 93 (1965), 273–299.
- [2] R. T. Rockafellar and R. J.-B. Wets, *Variational Analysis*, Springer, 1998.